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RESEARCH

# Comparative analysis of myoelectric activity and mandibular movement in healthy and nonpainful articular temporomandibular disorder subjects

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## Abstract

**Objectives** Temporomandibular disorders (TMDs) are one of the leading causes of craniofacial pain, and a high incidence of TMDs in young adults has been reported. Previous studies have used surface electromyography (sEMG) and mandibular kinematic analysis for TMDs diagnosis. This study aimed to provide normal reference values of sEMG and mandibular kinematics in Han Chinese adults, compare the sex differences and assess their diagnosis value on pain-free articular TMDs.

**Materials and methods** This observational study recruited healthy young adults with individual normal occlusion and nonpainful articular TMDs patients with disc displacement. The sEMG signals of the anterior temporalis (TA), masseter (MM), and sternocleidomastoid and digastric were recorded in the mandibular postural positions (MPP) and during maximal voluntary clenching (MVC) with K7 electromyograph. Mandibular kinematics, including maximum opening and opening/closing velocities, were assessed by K7 kinesiograph.

**Results** The sEMG reference values ranged from 3.0 to 4.3 μV in the MPP ($n=90$) in healthy subjects. The sEMG values during MVC, maximum opening, and opening velocity were significantly higher in males than in females. Also, these indices showed good diagnostic efficiency for pain-free articular TMDs ($n=26$).

**Conclusions** Reference values and sex differences in sEMG and mandibular kinematics in healthy subjects were determined. Compared with them, articular TMDs patients showed significantly lower myoelectric activity during maximal biting and restricted mouth opening range and velocity, which may assist in TMDs diagnosis.

**Clinical relevance** By analyzing sEMG of masticatory muscle and mandibular kinematics data from healthy Han Chinese young adults and TMDs patients with disc displacement, this study evaluated the sex difference and their diagnosis value on nonpainful TMDs with disc displacement.

**Clinical Trial Registration ID** NCT06372769.

**Keywords** Temporomandibular joint disorder · Surface electromyography · Mandibular kinematics · Sex difference · Diagnosis

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## Introduction

Temporomandibular disorders (TMDs) constitute a group of symptoms that involve pain and tenderness in the masticatory muscles or temporomandibular joint (TMJ), limitation of jaw movement, and joint noise during mouth opening/closing [1]. A high morbidity rate of TMDs was reported, in which approximately 31% of adults/elderly and 11% of children/adolescents were prone to have one or more symptoms [2].

Surface electromyography (sEMG) has been used in studying functional jaw muscle anatomy since the 1950s [3]. It is an objective and accurate tool to assist in muscular dysfunction diagnosis than palpation and visual observation [4]. Healthy people can relax jaw muscles at rest and tense muscle groups in a coordinated way, while the abnormal electromyographic signal can be collected from TMDs patients with muscle pain/dysfunction [5]. Further, the asymmetric sEMG activity also offers a valuable reference for diagnosing TMDs and evaluating the therapeutic effect of TMDs treatment [6, 7].

In addition, the association between mandibular kinematics and muscular diagnoses has been studied, and approximately 97% of TMDs patients with mandibular dysfunction have at least one muscular diagnosis, including protective co-contraction, myofascial pain, myospasm, or myalgia [8]. To our knowledge, mandibular kinematic values are also related to specific articular diagnoses. For example, abnormal mouth opening, obvious deviation, limited mandibular maximum excursions, and bony end-feel may be observed in TMDs patients with disc displacement, subluxation, or retrodiscitis [9]. In brief, muscle or joint symptoms directly or indirectly affect normal mandibular movement [10].

Establishing normal reference values of sEMG activity and jaw kinematic parameters is a significant step toward diagnosing and treating TMDs [11, 12]. It was found that the susceptibility of certain joint derangements or the awareness of pain differs among ethnic groups, and such differences refer to a genetic difference in the stomatognathic system that affects the prevalence of TMDs [13]. A recent study reported TMD prevalence in Italian adolescents (European, 29.7%) was obviously higher than that in Venezuelan adolescents (Hispanic, 17.9%), involving greater muscle pain and joint noise [14]. On one hand, there is a lack of quantitative and comprehensive analysis of sEMG activity and mandibular movement in healthy Han Chinese young adults, the effect of gender was also controversial. On the other hand, whether these indices would help the diagnosis of non-painful articular TMDs remains unclear.

The high-risk age group of TMDs and pain-free TMDs patients with disc displacement were recruited in this observational study. The non-invasive measurements of sEMG and mandibular movement made it possible to provide certain clinical references for TMDs diagnosis and explore the sex differences in these indices. The hypothesis was sEMG activity and mandibular kinematics had a good diagnostic efficiency for TMDs, and their normal values in healthy subjects were significantly affected by sex.

## Materials and methods

### Participants

Healthy volunteers were recruited from college students at Fujian Medical University who fulfilled the following criteria: (1) centred midline and no marked restriction and deviation of mouth opening and closing; (2) overjet and overbite of 1–3 mm, bilateral molar support with molar and cusp relation of Angle's class I, and (3) presence of complete permanent dentition, except third molars. They were excluded if they met one or more of the following exclusion criteria: (1) history of local or general trauma; (2) presence of systemic diseases, neurological or psychiatric disorders, muscular diseases, cervical pain, or TMDs based on the Diagnostic Criteria (DC/TMD) [15]; (3) pregnancy; (4) consumption of anti-inflammatory, analgesic, antidepressant, or myorelaxant drugs; (5) presence of parafunctional facets and anamnesis of parafunctional tooth clenching, bruxism, or unilateral chewing; (6) presence of obvious dentition crowding or spacing, malposed, supernumerary or fractured tooth, visible caries, tooth abrasion/hypersensitivity, toothache, periodontal disease, or occlusal discomfort; (7) fixed or removable restorations, tooth filling, or occlusal adjustment that affected the occlusal surfaces; and (8) previous or concurrent orthodontic, orthognathic, or TMJ treatment.

TMDs participants were recruited through public invitations, via recruitment posters and personal contacts. TMD diagnosis was based on DC/TMD Axis I protocol, and the symptom questionnaire was employed, followed by a pain-related interview, examiner commands, and CT/MRI (if necessary) [15]. Specifically, patients with unilateral or bilateral TMJ disc displacement symptoms, such as TMJ clicking, TMJ locking, and limitation in opening the mouth, were included in the study. The study did not cover the patients with myofascial pain, arthralgia, osteoarthritis, osteoarthrosis, TMJ fracture, dentofacial deformity, systemic disease affecting TMJ, and those who had undergone TMDs treatments. Other exclusion criteria for TMDs patients were the same with those of healthy volunteers. All subjects were Han Chinese people, and the recruitment and tests took place from February 2022 to July 2022. Envelopes were randomly allocated with numbers before the test,

and the testers and recorders had no information about the group to which they belonged.

## Surface electromyography

The sEMG of the anterior temporalis (TA), masseter (MM), sternocleidomastoid (cervical group, CG), and digastric (DA) were recorded simultaneously with an sEMG device (K7/EMG, Myotronics-Noromed, Inc., Tukwila WA, USA) using disposable silver/silver chloride bipolar surface electrodes (Duo-Trode, Myotronics-Noromed, Inc., Tukwila WA, USA).

The tests were conducted in a quiet environment, with the subjects comfortably seated on a fixed chair with a straight back, closed eyes, and relaxed head position. The electrodes were located by palpation of muscle contraction (Fig. 1A), and the ground electrodes were positioned on the subject's clavicle bone to provide a common reference to the differential input of the amplifier [16]. Each recording site was thoroughly cleaned and dried, and each pair of electrodes was placed on the site over the belly of the muscle and parallel with the muscle fibers. A conductive paste was used for good electrical contact with the skin. The electrode noise test was conducted to ensure the proper connection of the K7 EMG ground wire to the patient ground electrode, proper skin cleansing, quality of electrodes and electrode leads, and low electrical noise caused by an electric motor. The sEMG recording was started only when the software (K7 Program 15.0, Myotronics-Noromed, Inc., Tukwila WA, USA) revealed the absence of noise.

The raw sEMG data were recorded with a sampling frequency of 2000 Hz, and all recorded signals were band-pass

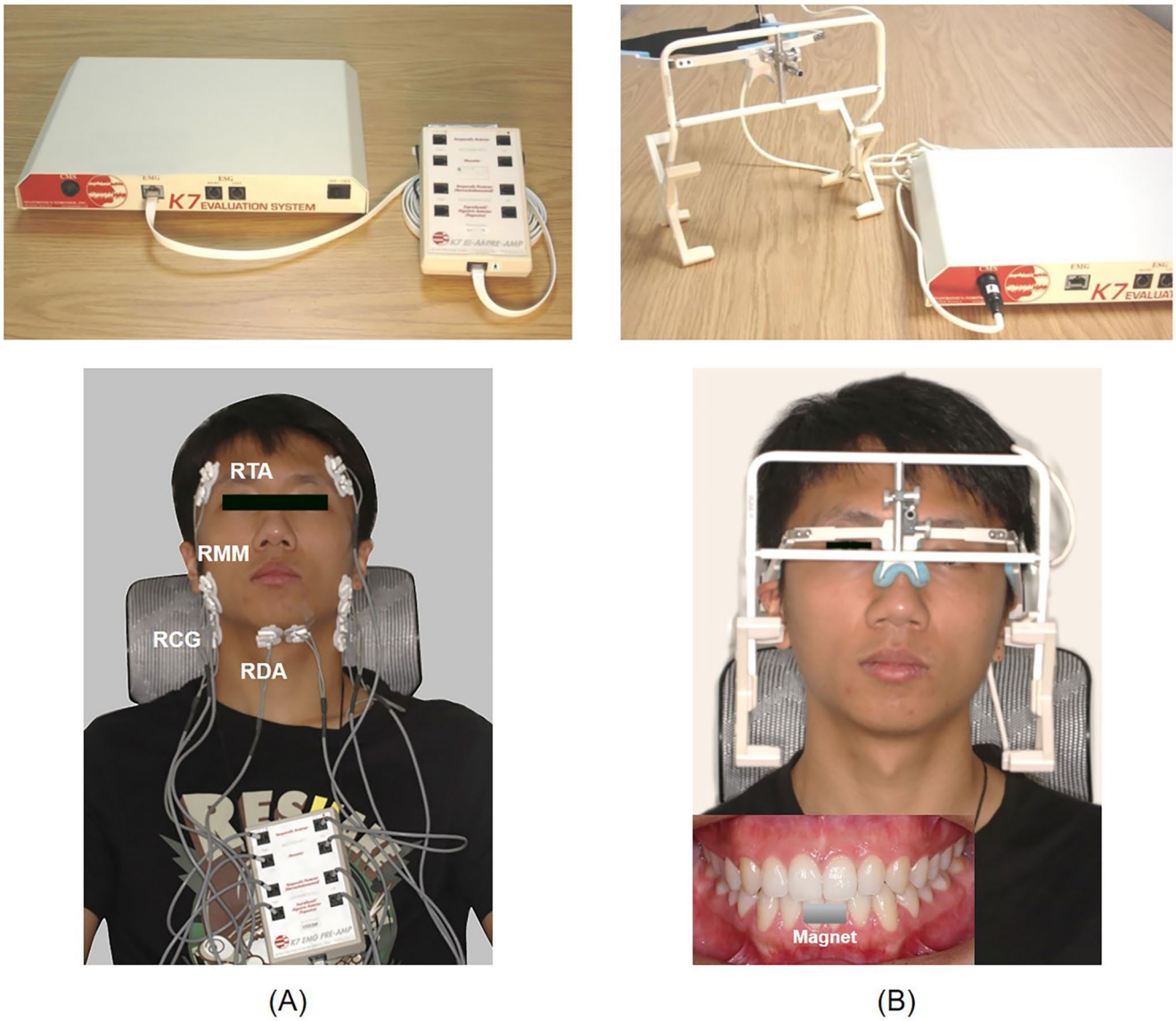


**Fig. 1** K7 surface electromyography and mandibular kinesiograph

filtered with a high- and low-pass Hamming filter with cut-off frequencies of, respectively, 15 and 650 Hz, and an additional 60 Hz 110 dB/decade notch filter. Three consecutive tracks were recorded, with a duration of 15 s and without any voluntary movement. The root mean square (RMS) value of sEMG (in μV) for each channel was displayed at the end of the tracing. First, sEMG was conducted on four pairs of muscles (TA, MM, CG, and DA) in the mandibular postural positions (MPP), and subjects were guided to maintain the face and jaw as relaxed as possible [17]. The relative resting activity of the masticatory/cervical muscles and the motor unit activity were averaged over time to quantify the amount of electrical activity.

To determine the relative efficiency of muscle function, the second test was performed to measure the sEMG of TA and MM during the maximal biting force against natural dentition (TA-ND and MM-ND) or cotton rolls (TA-CR and MM-CR). The disposable cotton rolls were used as standard food substitution with a length of 3.0 cm and a diameter of 1.0 mm. Each subject was asked to clench twice with maximum force, each held for 2.0 s, followed by a relaxation period of 1.0 s. This sequence was repeated after two cotton rolls were placed on the posterior teeth.

In the third test, subjects were guided to clench three times to monitor early motor unit recruitment as they closed from rest (MPP) through freeway space to initial tooth contact. Partial asymmetry indices (PAI) at the peak sEMG values of TA and MM (pTA and pMM) were automatically generated by software (Fig. 2B), and total asymmetry indices (TAI) during maximal voluntary clenching (MVC) were calculated using the following formula to determine muscle balance [11, 12]. For PAI and TAI, a zero value reflects similar muscle activity on the left and right sides.

$$PAI_{TA} = \frac{|pTA_{left} - pTA_{right}|}{\mathrm{p}TA_{right}}\%$$

$$PAI_{MM} = \frac{|pMM_{left} - pMM_{right}|}{\mathrm{p}MM_{right}}\%$$

$$TAI = \frac{|TA_{right} + MM_{right} - TA_{left} - MM_{left}|}{TA_{right} + MM_{right} + TA_{left} + MM_{left}}\%$$

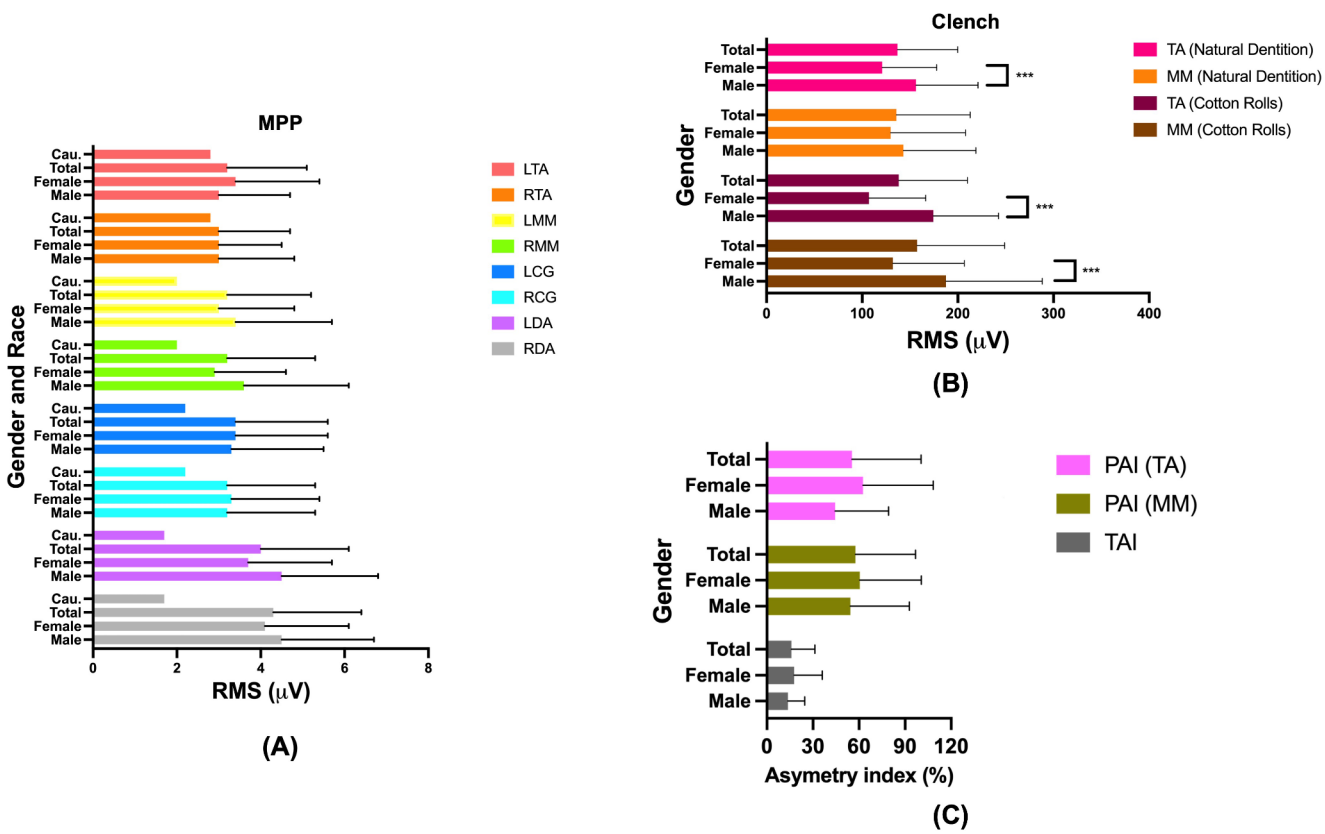


**Fig. 2** (**A**) The maximum vertical opening, maximum anteroposterior movement from centric occlusion, maximum lateral deviation, and straight-line distance from centric occlusion to maximum opening; (**B**) The maximum opening velocity, average opening velocity, maximum closing velocity, average closing velocity, and maximum velocity of terminal tooth contact

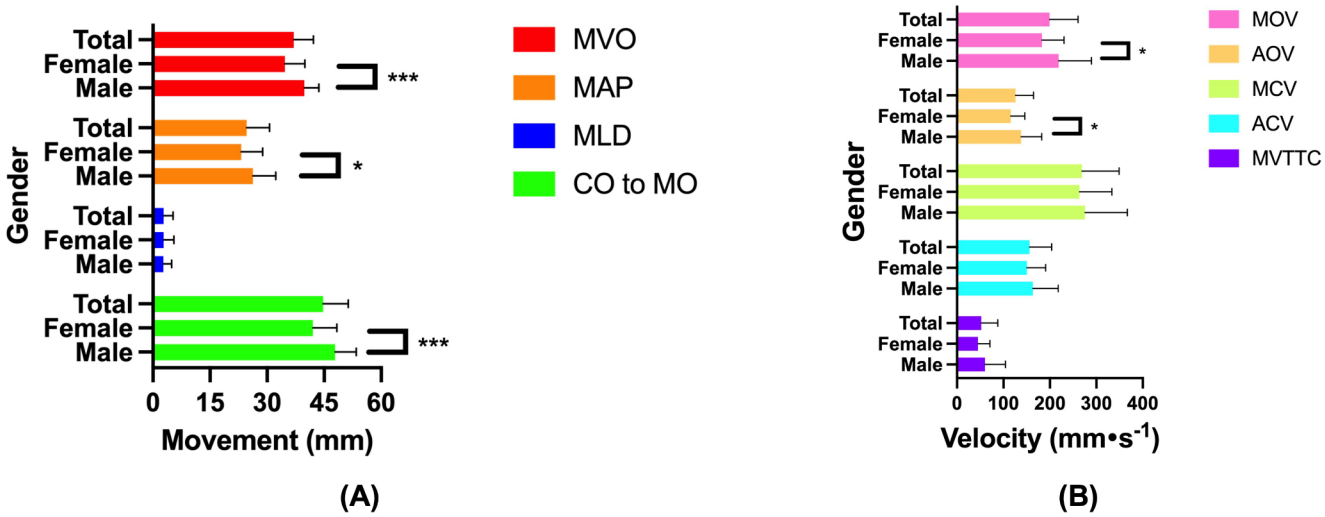


**Fig. 3** (**A**) The average sEMG values for all tested muscles. Cau represents the normal values in Caucasians and Total is the normal values in male and female Han Chinese; (**B**) The bilateral sEMG of the temporalis anterior and masseter during maximal voluntary clenching against natural dentition and cotton rolls; (**C**) The asymmetry index of the temporalis anterior and masseter during maximal voluntary clenching against natural dentition

## Mandibular kinematics

Kinesiographic recordings were performed using a kinesiograph (K7/CMS, Myotronics-Noromed, Inc., Tukwila WA, USA) that measured mandibular movements with an accuracy of 0.1 mm in the vertical, anteroposterior (AP), and lateral axes. An array with eight magnetic sensors mounted on the subject's head tracked the motion of the small magnet attached to the labial surface of the lower central incisor at a sample rate of 125 Hz (Fig. 1B). The maximum mouth opening (MMO) and opening and closing velocities were measured.

The first tracing was a simultaneous sagittal and frontal trace of the mandibular trajectory traversed during mouth opening and closing. Each subject was asked to open their mouth as wide as possible and then close it. Then, they opened and closed their mouths, starting from the intercuspal position (ICP) and moving back as fast as possible. The sagittal, frontal, and velocity tracings were simultaneously recorded. Meanwhile, the maximum velocity of terminal tooth contact (MVTTC) was measured, defined as the closing speed at 0.8 mm below the centric occlusion (CO).

All measurements were repeated twice, with a gap of 15 min between two recordings, and the average values were calculated. Additionally, each participant was assessed three times by different examiners at an interval of three days. Reliability was calculated using the intraclass correlation coefficient (ICC). The inter- and intra-rater reliability was calculated with ICC using the average data of each test per session.

## Statistical analysis

We conducted a power analysis in this study to determine the appropriate sample size. The parameters for the power analysis included the effect size of Cohen's f (0.25), an alpha level of 0.05, and a desired power of 0.80. GPower 3.1.9 software (OARC, UCLA, USA) indicated that a total sample size of 30 participants would be necessary to achieve sufficient power for detecting the expected effect. Statistical analysis was performed using SPSS 27.0 software (IBM, NY, USA). The level of significance was set at $P \leq 0.05$ for all tests. Shapiro-Wilk and Levene tests were used to assess the data's normality and variance homogeneity, respectively. Significant differences in sEMG values and mandibular kinematics between sex and between healthy and TMDs subjects were established by the analysis of variance (ANOVA). The Mann-Whitney U test would be used if there is any non-parametric data. Further, the receiver operator characteristic (ROC) curve was used to analyse their diagnostic efficiency for TMDs, and their effect on the severity of TMDs (included healthy, ADDwR and ADDwoR levels. ADDwoR served as a more severe subtype than ADDwR in this study) was also investigated using Spearman correlation.

## Results

A total of 436 college students participated in the initial screening, and 90 healthy volunteers were finally recruited in this study (49 women and 41 men; mean age 20.7 ± 0.7 years). Meanwhile, 26 nonpainful articular TMDs patients

were recruited based on the above-mentioned criteria (19 women and 7 men; mean age 29.7 ± 4.8 years). They were further divided into anterior disk displacement with reduction (ADDwR) group ($n = 13$) and anterior disc displacement without reduction (ADDwoR) group ($n = 13$) by MRI.

Intra-interrater reliability showed high reliability (above 0.85) for each subject. In healthy subjects, there was no significant sex difference in the bilateral sEMG of the TA (L: $P = 0.19$, R: $P = 0.99$), MM (L: $P = 0.81$, R: $P = 0.54$), CG (L: $P = 0.98$, R: $P = 0.77$), and DA (L: $P = 0.05$, R: $P = 0.63$) in the MPP (Fig. 2A).

For the bilateral TA, significantly higher average sEMG values were found in males than in females during MVC with and without cotton rolls ($P < 0.001$) (Fig. 2B). Male subjects also had higher sEMG activity of the bilateral MM when clenched against cotton rolls ($P < 0.001$). The muscle asymmetry indices during MVC. The $PAI_{TA}$ ranged from 46.1 to 64.7%, $PAI_{MM}$ ranged from 49.6 to 65.8%, and TAI ranged from 13.1 to 15.6% in healthy subjects. No difference in PAI (TA: $P = 0.09$, MM: $P = 0.52$), and TAI ($P = 0.24$) was found among sexes (Fig. 2C).

Figure 3A and B present the range of jaw movements during MMO and the quick mouth opening/closing velocity. Higher values of maximum vertical opening (MVO) ($P < 0.001$), maximum AP movement from CO (MAP) ($P < 0.05$), and linear distance from CO to maximum opening (CO-MO) ($P < 0.001$) were found in males than in females. Additionally, the male subjects exhibited a faster opening speed according to the maximum opening velocity (MOV) and average opening velocity (AOV) ($P < 0.05$).

In Fig. 4, although there was no significant difference in sEMG value of MPP and asymmetry index among healthy subjects and ADDwR and ADDwoR patients, statistically higher sEMG activity of TA and MM were recorded in healthy subjects during MVC against natural dentition and cotton rolls ($P < 0.001$). Furthermore, ADDwR patients showed higher sEMG when clenched against cotton rolls than that in ADDwoR patients ($P < 0.05$). Moreover, healthy subjects exhibited a larger jaw movement range (MVO and CO-MO) and faster movement speed (MOV, AOV, MCV, and ACV) than those of TMDs patients.

For ROC analysis, the area under the curve (AUC) of MCV (0.95), MVO (0.90), and TA-ND (0.86) were larger than other indices. The optimal MCV cut-off level was 190.2 mm/s, with a sensitivity and specificity of 100.0% and 82.7%, respectively. Meanwhile, MVO showed an optimal cut-off at 30.6 mm with a sensitivity and specificity of 100.0% and 90.2%, respectively (Fig. 5).

Spearman correlation analysis showed that the severity of TMDs (Healthy = 1, ADDwR = 2, ADDwoR = 3) were strongly correlated with MCV ($r_s$=-0.6260, $P < 0.001$) and MVO ($r_s$=-0.6385, $P < 0.001$) (Table 1). Additionally, there was a moderate correlation between TA-ND and the severity of TMDs ($r_s$=-0.4843, $P < 0.001$).

## Discussion

This study attempts to estimate reference values of sEMG activity of the masticatory muscle and mandibular kinematics in healthy Han Chinese young adults with normal occlusion and evaluate their diagnosis efficiency on pain-free articular TMDs. The null hypothesis was supported by the results that sex remarkably influenced the sEMG values during MVC, mouth opening, and opening velocity, which also showed good diagnostic value for TMDs with disc displacement.

sEMG is a non-invasive tool for gathering electrical signals produced during muscular contractions, and the quantitative analysis of sEMG has become a supplementary diagnostic method for TMDs in recent decades [3]. Determining normal parameters provides an important reference for the health assessment of TMJ. However, there is a lack of relevant data on Han Chinese, the largest ethnic group in China. We found that the rest sEMG in Han Chinese may be higher than that in Caucasians, especially in MM, CG, and DA. In a previous study, Ferrario et al. [12] measured the average sEMG activity of TA and MM in the MPP and ICP, and during MVC in 92 healthy Caucasian young adults with natural dentition. Similarly, Campillo et al. [11] aimed to establish normal reference values of sEMG in Spanish young adults with ideal dentition. In the present work, the average sEMG activity values in the MPP and during MVC were slightly lower than in previous studies. Furthermore, 125 μV was regarded as the minimum threshold for physiological clenches by Cooper et al. [18], who collected bioelectronic testing data from dentists in nine countries. Although strict inclusion criteria for the participants were used, variations in experimental methodologies and population features remained among studies, including sample size, test equipment, ethnic group, and dietary habits.

TMDs may cause chronic facial pain besides headaches and cervical spine disorders. The craniomandibular region and upper cervical spine are closely related in anatomy, biomechanics, and neurophysiology [19]. There is a functional link between the masticatory and cervical muscles based on a trigeminal-cervical reflex [20]. The TMDs patients with myalgia were reported to have higher resting sEMG activity of the sternocleidomastoid and trapezius muscles than that of healthy subjects [21]. However, only the TMDs patients without pain in the facial muscle were enrolled in this study, which may explain why they did not show a significant difference in sEMG value in the MPP compared with healthy subjects. Nevertheless, TMD patients with disc

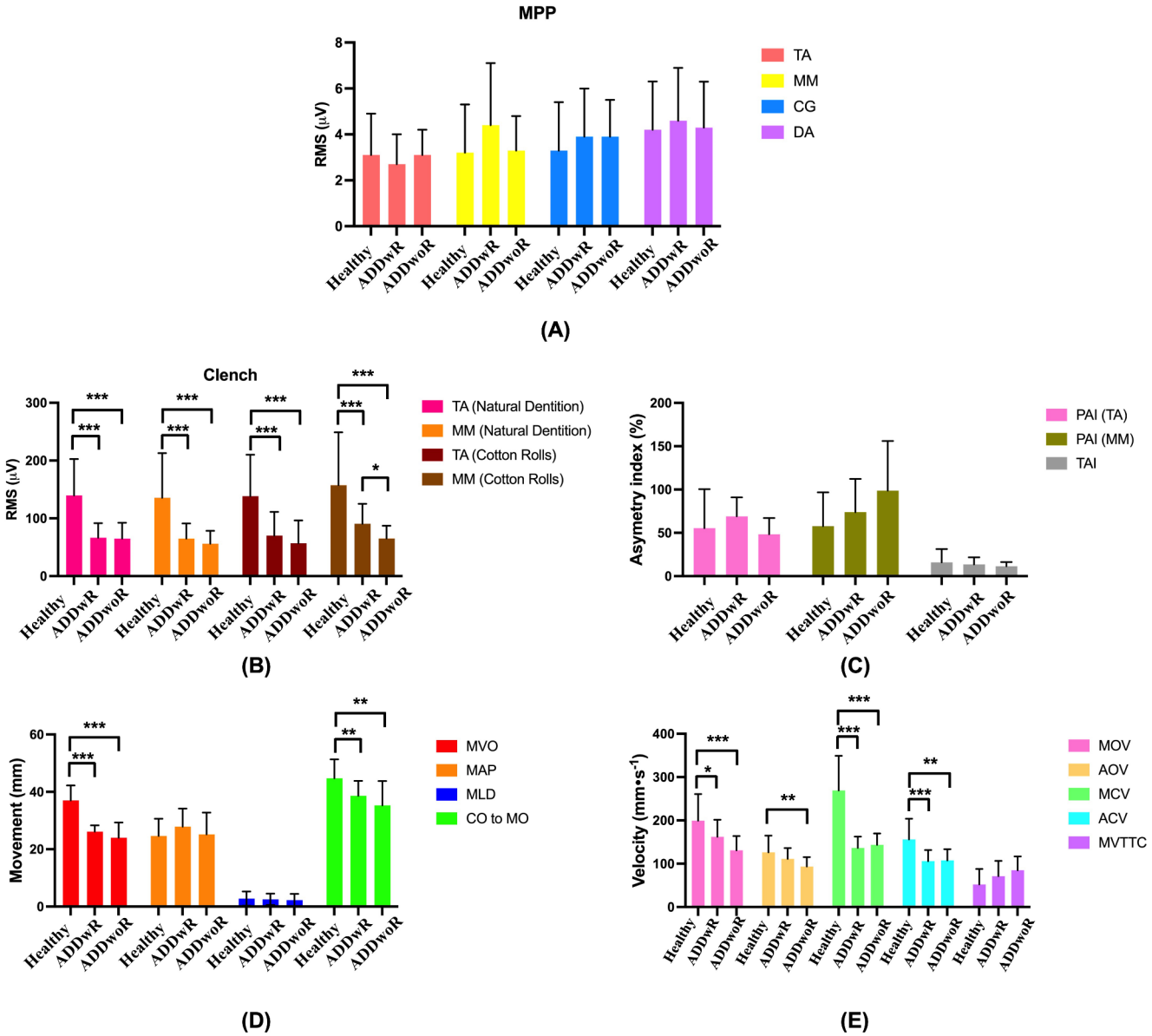


**Fig. 4** (**A**) The bilateral sEMG of the temporalis anterior, masseter, sternocleidomastoid, and digastric in the mandibular postural position in healthy, anterior disc displacement with reduction (ADDwR) and anterior disc displacement without reduction (ADDwoR) subjects; (**B**) The bilateral sEMG of the temporalis anterior and masseter during maximal voluntary clenching against natural dentition and cotton rolls in healthy, ADDwR, and ADDwoR subjects; (**C**) The asymmetry index of the temporalis anterior and masseter during maximal voluntary clenching against natural dentition in healthy, ADDwR, and ADDwoR subjects; (**D**) The maximum vertical opening, maximum anteroposterior movement from centric occlusion, maximum lateral deviation, and straight-line distance from centric occlusion to maximum opening in healthy, ADDwR, and ADDwoR subjects; (**E**) The maximum opening velocity, average opening velocity, maximum closing velocity, average closing velocity, and maximum velocity of terminal tooth contact in healthy, ADDwR, and ADDwoR subjects

displacement and/or muscle disorders should be included in the further study.

In addition, significantly higher sEMG activity was recorded during MVC in males than in females. Fogle et al. [22] also found a significant association between sEMG data and sex while biting rather than resting. This can be explained by the sex differences in muscle recruitment, muscle thickness, and craniofacial morphology, and males were likely to have a higher proportion of thick type II muscle fibres and greater facial length than females [23–25].

Moreover, previous studies have shown that patients with functional disturbances in the masticatory system present decreased bite force compared to healthy individuals [26–29]. Although the most common cause of decreased muscle function reported accordingly is the pain of myogenic and/or arthritis, the reduced muscle activity in non-painful TMDs

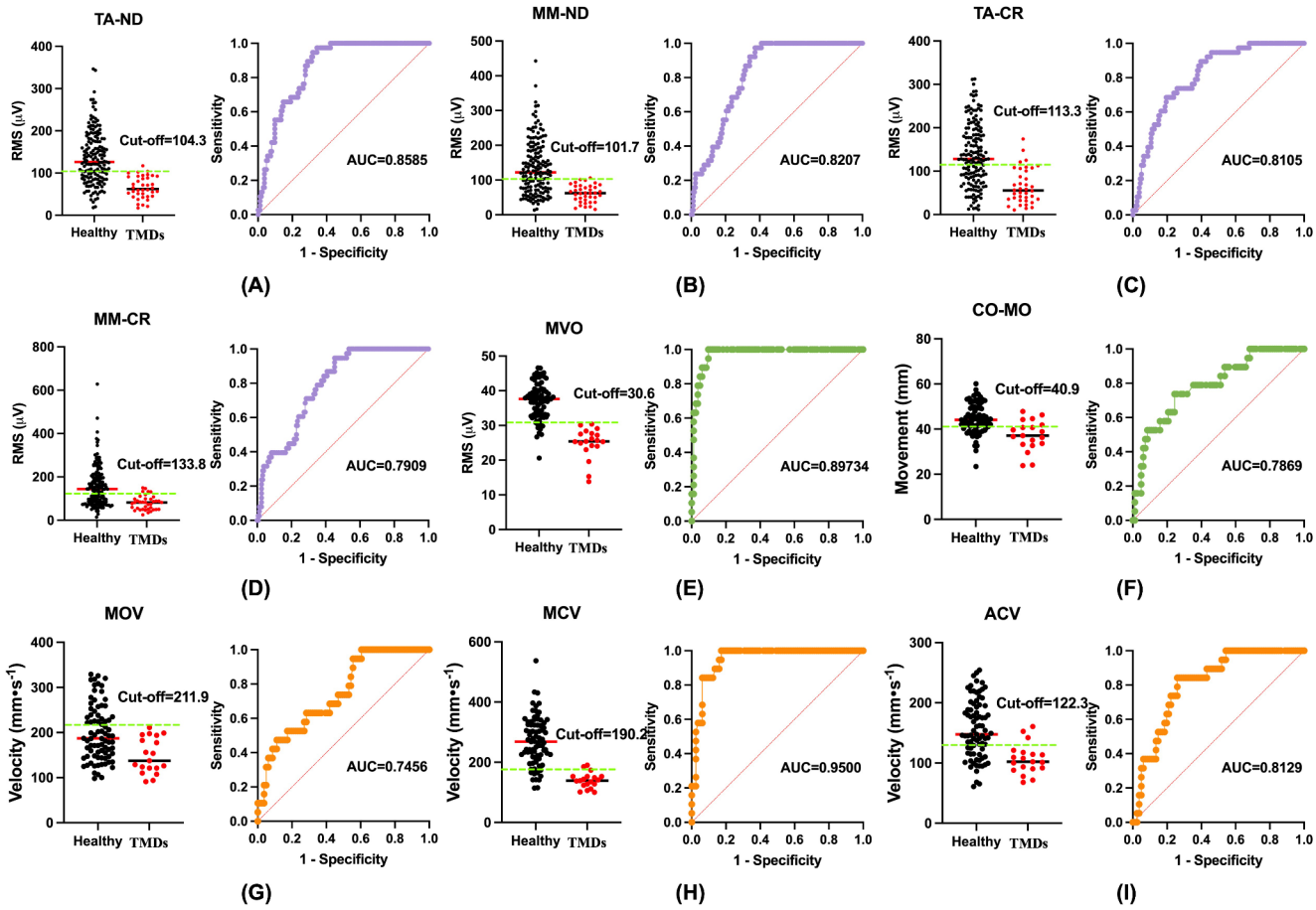


**Fig. 5** Receiver operating characteristics (ROC) curve for sEMG and mandibular kinematics. Diagnostic efficacy of those values for Temporomandibular Disorders (TMDs) was assessed using area under the curve (AUC) and cut-off point. The bilateral sEMG of (**A**) the temporalis anterior and (**B**) masseter during maximal voluntary clenching against natural dentition; The bilateral sEMG of (**C**) the temporalis anterior and (**D**) masseter during maximal voluntary clenching against cotton rolls; (**E**) The maximum vertical opening and (**F**) straight-line distance from centric occlusion to maximum opening; (**G**) The maximum opening velocity, (**H**) maximum closing velocity, and (**I**) average closing velocity

patients may be explained by a protective activity of elevator muscles and progressive alterations in the metabolism of energy in the muscle after a chronic fatigue under high pressure. Also, Ow et al. [30] found that patients with TMDs demonstrate significant variations in the values of maximal bite force at multiple repeated measures, which indicates the loss of sensory acuity in regulating the masticatory muscle function. Also, the healthy subjects presented statistically higher sEMG values of TA and MM during MVC than those in TMDs patients in this work, and TA-ND showed a good diagnostic efficiency and a moderate correlation with the severity of TMDs that was similar to the results of Todic's [26] and Kogawa's [29] study.

Prior studies have noted the importance of asymmetrical muscle activity. Higher asymmetrical indices were seen in myogenous TMDs patients than in healthy participants, which would be improved by an effective treatment with occlusion splints [6, 7]. The asymmetrical muscle activity and the remission of symptoms may be caused by unstable occlusion, bringing about more prolonged pressure to the TMJ and tissue ischemia [6]. In this work, $PAI_{MM}$ was higher than $PAI_{TA}$, which may be attributed to the larger cross-sectional area and more muscle fibres of MM than TA [31]. Meanwhile, TAI ranged from 11.9 to 16.3% in healthy subjects, slightly larger than the results of previous studies and even TMDs patients without significant difference. Therefore, the diagnostic value of asymmetrical muscle activity should be careful consideration of the wide differences among individuals and patients with muscle disorders [32]. It is more appropriate to be used in self-pre- and post-control observation or combined with other diagnostic information.

Abnormal jaw movement is a prominent symptom of TMDs patients [8]. Limited jaw movement was reported to be associated with acute pain in the related structures of TMJ or the adaptable protection of the musculoskeletal

**Table 1** The Spearman correlation analysis of TMDs with sEMG signal and mandibular kinetics

| sEMG signal and mandibular kinetics | | The severity of TMD (Healthy = 1, ADDwR = 2, ADDwoR = 3) |
|---|---|---|
| TA-ND | Spearman Correlation | -0.4843 |
| | Sig. (2-Tailed) | < 0.0001 |
| MM-ND | Spearman Correlation | -0.4392 |
| | Sig. (2-Tailed) | < 0.0001 |
| TA-CR | Spearman Correlation | -0.4267 |
| | Sig. (2-Tailed) | < 0.0001 |
| MM-CR | Spearman Correlation | -0.4080 |
| | Sig. (2-Tailed) | < 0.0001 |
| MVO | Spearman Correlation | -0.6385 |
| | Sig. (2-Tailed) | < 0.0001 |
| CO-MO | Spearman Correlation | -0.3896 |
| | Sig. (2-Tailed) | < 0.0001 |
| MOV | Spearman Correlation | -0.3465 |
| | Sig. (2-Tailed) | 0.0004 |
| MCV | Spearman Correlation | -0.6260 |
| | Sig. (2-Tailed) | < 0.0001 |

The severity of TMDs includes three levels: healthy, ADDwR, and ADDwoR. The anterior disc displacement without reduction (ADDwoR) served as a more severe subtype than the anterior disc displacement with reduction (ADDwR). TA-ND: sEMG of temporalis anterior during maximal voluntary clenching against natural dentition; MM-ND: sEMG of masseter during maximal voluntary clenching against natural dentition; TA-CR: sEMG of temporalis anterior during maximal voluntary clenching against cotton rolls; MM-CR: sEMG of masseter during maximal voluntary clenching against cotton rolls; MVO: Maximum vertical opening; CO-MO: Straight-line distance from centric occlusion to maximum opening; MOV: Maximum opening velocity; MCV: Maximum closing velocity

system against further injury in chronic TMDs patients [9, 10]. Although significantly larger MVO was observed in males than in females, the sex difference could be affected by many potential confounding factors, such as mandibular length [33], mandibular rotation angles [34], and the shapes of the condylar pathway [35]. Males with larger mandibles and rotation angle and longer pathways would have larger MVO than females. Moreover, TMDs patients with disc displacement showed remarkably lower MVO than healthy subjects. However, we did not find a significantly positive correlation between TMJ disc reduction ability and mouth opening as the previous study reported, possibly due to the inadequate sample size. Therefore, to further determine ADDwR or ADDwoR, kinesiograph should be combined with other techniques (e.g., computer vision) to assist clinicians when it is difficult to obtain information through MRI [36].

The significant correlation between movement velocity and intervertebral disc herniation had been described before [37], and the improvement in opening/closing velocity was used to evaluate the efficacy of myorelaxation therapy on TMDs [38]. There were significant differences in opening velocity for the types of muscle fibres between the sexes [39, 40], and males had a stronger lateral pterygoid muscle activity than females [41]. This can explain the reason for faster MOV and AOV in males than in females in healthy subjects. For diagnosis value, a significantly lower jaw movement velocity was recorded in TMDs patients than that in healthy volunteers, and the AUC of MCV was close to 1, indicating its good predictability for disc displacement. Also, there was a strong correlation between the severity of TMDs and MCV, which may offer a good reference value for diagnosing TMDs. This result was similar to the previous study, in which TMDs patients' closing velocity during gum chewing was less compared to their control subjects, and it can be ascribed to inefficient muscle recruitment and muscle fatigue [42, 43].

Lower MVTTC was found in patients with severe rather than mild periodontitis because of the protective co-contraction of antagonistic muscles derived from a neuromuscular feedback mechanism [44]. Although the MVTTC may provide the possibility of assessing the mandible motor function, there was no significant difference between healthy subjects and TMDs patients in this study. It may be related to the types of TMDs and sample size, and further investigation is clearly warranted.

Recording the sEMG and mandibular movement tracings with non-invasive, repeatable, and portable electronic devices is useful in assisting the clinical diagnosis of TMDs. Furthermore, the normal reference ranges and sex differences in sEMG activity and kinematics may provide important information about etiological analysis and lesion positioning. However, further research on a larger sample size needs to be carried out, and the limitations are related to the possible interferences that exist in the collection of sEMG and kinematic data. The psychological factors and muscle fatigue in subjects and the weight of the EMG preamp and magnetic sensors have a potential risk of bias.

## Conclusion

1. Reference values for sEMG of the jaw muscles and the mandible kinematics were determined in healthy Han Chinese young adults with normal occlusion.
2. Sex differences were evident in the sEMG of the anterior temporalis and masseter during maximal voluntary clenching. Significantly higher maximum vertical opening and opening velocity values were found in males than in females.
3. Maximal closing velocity, maximal vertical opening, and sEMG of the anterior temporalis during maximal

voluntary clenching may provide a good diagnostic value for pain-free temporomandibular joint disc displacement.

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**Author contributions** H.L. Lin designed the study. X.J. Xing, J.M. Peng and X.Y. Wu acquired, analysed and interpreted the data. X.J. Xing and Y.L. Cheng drafted the manuscript, and H. Cheng and H. Yu revised it critically for important intellectual content. All authors read and approved the final manuscript.

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**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflicts of interest** The authors declare that they have no conflict of interest.

**Ethical approval** The study was conducted by the Declaration of Helsinki and approved by the Ethics Review Committee for Biomedical Research of the School and Hospital of Stomatology, Fujian Medical University (Fujian, China), with the IRB number of FJMU 2021-056.

**Informed consent** Written informed consent was obtained from all individual participants included in this study. No identifying information is included in this article.

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